MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : PEC-CS702E/PEC-IT702F Cyber Security

UPID : 007558

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(I) What is trojan horse?

(II) Governments hired some highly skilled hackers for providing cyber security for the country or state. These types of hackers are termed as _____.

(III) _____ is a code injecting method used for attacking the database of a system / website.

(IV) Which of them is not a wireless attack?
a) Eavesdropping
b) MAC Spoofing
c) Wireless Hijacking
d) Phishing

(V) Which of these is NOT a scanning type?
a. Null Scan b. Xmas Tree Scan c. SYN Stealth d. Cloud Scan

(VI) Which malware does not clone or replicate itself through infection?

(VII) Which is the least strong security encryption standard?

(VIII) Which of the following is a component of cyber security?
a) Internet Of Things
b) AI
c) Database
d) Attacks

(IX) Which of the following is not a step followed by cyber-criminals in data breaching?
a) Exfiltration b) Research and info-gathering c) Attack the system d) Fixing the bugs

(X) Which of the following is defined as an attempt to harm, damage or cause threat to a system or network?
a) Digital crime
b) Threats
c) System hijacking
d) Cyber Attack

(XI) Which of the following is a type of cyber attack?
a) Phishing
b) SQL Injections
c) Password Attack
d) All of the above

(XII) Define CIA Triad.

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. (i)Define Hacker. List Out the Types of hackers
(ii)What is a Backdoor? [5]

3. What is the difference between IDS and IPS? [5]

4. Explain port scanning. [5]

5. Discuss the significance of intellectual property rights (IPR) in cyberspace. [5]

6. Explain brute force attack and the ways to prevent it. [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

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7. How can be Intrusion Detection system is the backbone of Information system? Justify along with its categories? What do you mean by Application Security? [15]

8. With a neat diagram, explain the zachman framework for enterprise architecture. Discuss primitive models versus composite models. [15]

9. (a) Explain the major stages of a forensic investigation process. [10]

   (b) Describe the role of auditing in cybersecurity and forensics. [5]

10. Explain the difference between Virus, Worms and Trozan Horse. What is the data security consideration? Explain in this reference Data backup security, Data archival security and Data disposal considerations. [15]

11. (a) Explain in detail the concept of social engineering. Describe its techniques and preventive measures. [10]

    (b) What do you mean by SQL injection attacks. [5]

*** END OF PAPER ***

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